System Design

Engineering Design Rationale

Described an overview of the vehicle as an overall system and its subsystems
Described the engineering reasoning behind the vehicle systems and components
Data was used to compare and select from among alternative designs/tooling

Hello I am Mohammed Badri, Chief technical Officer of Phoenix Robotics, and today, I'm thrilled to walk you through the intricate subsystems of our ROV. Let's embark on this journey together, exploring the design choices and engineering reasoning that shape our underwater machine.

Firstly, let's delve into the Thrusters. This year, we opted for the Blue Robotics T-200's renowned for their exceptional efficiency. With a four-motor configuration, we prioritized efficiency to navigate underwater terrains effectively. Control-wise, we've integrated Electronic Speed Controllers (ESCs), programmed via Arduino Mega, providing seamless control. Our user interface consists of a pair of joysticks, delivering analog signals to the Arduino, which then commands the ESCs, ensuring precise motor responses.

Now, let's shift our focus to the Manipulator subsystem—a critical component tailored to meet the demands of various tasks. After careful consideration, we selected two analog servos—a 360-degree and a 270-degree variant—to facilitate pivoting movements. These servos power our claw mechanism, enabling both opening and closing actions, alongside rotation control. Opting for an analog system was a strategic decision, backed by rigorous testing that demonstrated superior stability without compromising performance.

Moving on to the Cameras subsystem, we've implemented a dual-camera setup to fulfill diverse task requirements while keeping costs in check. Our primary system comprises four analog cameras strategically positioned to provide comprehensive coverage. Additionally, for specialized tasks like 3D coral reef modeling, we've incorporated a secondary digital camera connected to a laptop with specialized software. This dual-camera approach not only ensures task-specific functionality but also optimizes cost efficiency by leveraging analog systems' performance benefits.

Temperature monitoring was part a of task 2 leading us to integrate a dedicated Temperature Sensor subsystem. We meticulously evaluated various options, prioritizing accuracy, reliability, and durability. The selected sensor delivers precise temperature readings, ensuring data accuracy in our tasks.

In conclusion, every aspect of our ROV's design has been carefully considered, with a focus on efficiency, performance, and cost-effectiveness. By leveraging data-driven insights, we've crafted

a robust and versatile underwater vehicle poised to tackle exploration challenges with unparalleled precision. Thank you for your attention, and I'm eager to address any questions you may have.